Solution

GRAVITY-05

Class 09 - Science

Section A

1.
   (d) Increases

   Explanation: At Antarctica, the value of acceleration due to gravity g is more, therefore, the weight of sugar bag will increase. We know at the equator, acceleration due to gravity is minimum and at the pole (Antarctica is at the South Pole), it is maximum. Therefore, the weight of the bag increases in Antarctica.

2. **(a)** Buoyant force

   Explanation: The upward force experienced by the body when submerged in a fluid is called buoyant force. Pressure increases with depth as a result of the weight of the overlying fluid. So, the pressure at the bottom of an object submerged in a fluid is greater at the top of the object. This pressure difference results in a net upwards force on the object. Buoyant force depends upon nature of fluid and density of substance.

3. **(a)** all of these

   Explanation: The amount of heat energy lost by the hot body in a given time depends on temperature, surface nature and the area affect the loss.

4.
   (d) 0.83

   Explanation: According to Archimedes's principle, buoyant force, FB = Volume of the object immersed in liquid x density of liquid x acceleration due to gravity Let V be the volume of object.

   For water: $F_B = \frac{3V}{4} \times \rho_w \times g = \frac{15}{2} V \rho_w$ ...(i)

   For alcohol: $F_B = \frac{9\text{ V}}{10} \times \rho_a \times g = 9\text{ V}\rho_a$ ...(ii)

   From eqn. (i) and (ii),

   $\frac{15}{2} V \rho_w = 9 V \rho_a \therefore \frac{\rho_a}{\rho_w} = \frac{15}{2 \times 9} = \frac{15}{18} = 0.83$

5.
   (b) R

   Explanation: Weight of passenger on Earth = m $\times$ g_e = 60 $\times$ 10 = 600 N

   Weight of passenger on Mars = m $\times$ g_m = 60 $\times$ 4 = 240 N

   As the passenger is going from the Earth to the Mars in a spaceship moving with a constant velocity so the acceleration due to gravity because of Earth and Mars becomes negligible (zero) at some point in space. Thus, weight of passenger becomes zero. After sometime, as he reaches near to the Mars, his weight increases because of acceleration due to gravity of Mars. Finally, at the surface of Mars, his weight becomes 240 N.

6.
   (c) U = W_a - W_l

   Explanation: Up thrust = apparent lost in weight of the object

   Up thrust = W_a - W_l

7.
   (c) 2.45 ms^{-2}

   Explanation: g is inversely proportional to the square of the distance between the earth and the body. As the distance gets twice the g will become one - fourth.

8. **(a)** is independent of mass and radius of the earth

   Explanation: The value of g is independent of the mass of the object and only dependent upon location - the planet the object is on and the distance from the center of that planet.

9.
   (b) Statement 1 is false but statement 2 is true.

   Explanation: Inside the satellite, g = 0

$T=2\pi\sqrt{\frac{I}{g}}=2\pi\sqrt{\frac{1}{0}}=\infty$

Thus, the pendulum will not oscillate.

10.

(b) 2.5 g wt

Explanation: L.C. = $\frac{25-0}{10}$ = $\frac{25}{10}$ = 2.5 g wt

11.

(d) 44.2 ms^{-1}

Explanation: V^2 - u^2 = 2gs

u = 0 g , g = 10ms^{-1}, s = 100 m

V^2 = 2 $\times$ 10 $\times$ 100 = 2000

V = 44.2 m/s

12.

(d) gravity

Explanation: Earth's atmosphere is the layer of gases around the Earth. The atmosphere is held in place by Earth's gravity.

13. **(a)** Difference in density of the body and half the density of the liquid.

Explanation: Weight = W - U

$V\rho g\ -\ \frac{V}{2}\sigma g\ =\ Vg\ \left(\rho\ -\ \frac{\sigma}{2}\ \right)$

14. **(a)** 10 kg

Explanation: Gravity on the Moon has approximately 1/6th of the strength of gravity on Earth, a man who weighs 60kg on Earth would weigh approximately 10kg on the Moon. His mass, however, remains constant.

15.

(b) All of these

Explanation: Density, mass and radius of moon is different than earth so the acceleration due to gravity is different and calculatively obtained 1/6th of Earth's gravity.

16.

(b) In the direction opposite to the direction of motion.

Explanation: When a ball is thrown then the direction of motion of the ball and vertically upwards then the direction of motion velocity is the same, i.e. upward of the ball is upwards but the acceleration due to gravity acts downwards towards the earth. Thus, the direction of acceleration is opposite to the direction of motion of the ball.

17.

(c) 78.4 m

Explanation: S = ut + $\frac{1}{2}$ gt^2 where S = distance, U = initial velocity, t = time and g = acceleration due to gravity.

Given: U = 0 m/sec, t = 4sec.

we know g= 9.8 m/sec^2

Therefore, S = $\frac{1}{2}gt^2$

S = $\frac{1}{2}$ $\times$ 9.8$\times$ 16

S = 78.4 meter.

18. **(a)** thrust

Explanation: Force exerted by an object perpendicular to the surface is called thrust.

19.

(c) 2.17 N

Explanation: Here, M_p = 2 M_e and R_p = $3R_e$

m = 1 kg, W_p = ?

As $g_e=\frac{GM_e}{R_e^2}$ and $g_p=\frac{GM_p}{R_p^2}$

$\frac{g_p}{g_e}=\frac{M_p}{R_p^2}\times\frac{R_e^2}{M_e}$

or$g_p = g_e \times \frac{M_p}{M_e} \times \left(\frac{R_e}{R_p}\right)^2$

or$g_p = 9.8 \times 2 \times \left(\frac{1}{3}\right)^2$ = 2.17 m/s^2

$W_p = mg_p = 1 \times 2.17 = 2.17$ N

20.

(d) $g \propto \frac{1}{r^2}$

Explanation: g = $\frac{GM}{r^2}$

So, g $\propto$ M and g $\propto \frac{1}{r^2}$

21.

(b) All of these

Explanation: Acceleration due to gravity varies with height, depth and shape of the planet as follows-

- As the height (altitude) increases acceleration due to gravity (g) decreases.
- As the depth increases acceleration due to gravity (g) decreases.
- Acceleration due to gravity is inversely proportional to the radius.

22. **(a)** Both statement 1 and statement 2 are true but statement 2 is not the correct explanation of statement 1.

Explanation: Variation of g with depth from surface of earth is given by $g' = g\left(1 - \frac{d}{R}\right)$

At the centre of earth, d = R, $\therefore g' = g\left(1 - \frac{R}{R}\right) = 0$

$\therefore$ Apparent weight of body = mg' = 0

23.

(d) varying velocity

Explanation: The Earth's orbit is not perfectly circular. As a result its velocity changes slightly over the course of the year.

24.

(d) it is completely immersed in the liquid.

Explanation: When a solid is completely immersed in the liquid, there is a maximum apparent loss in its weight due to the maximum volume of liquid displaced.

25.

(c) $P_A = P_B$

Explanation: Pressure varies with height of liquid in a container and height is same for both the points so pressure at both points remain same.

26. **(a)** $\frac{24}{25}h$ m

Explanation: Here, height = h m, time =$\frac{T}{5}$ s

As s = $ut + \frac{1}{2}at^2$ $\therefore h' = \frac{1}{2}g\left(\frac{T}{5}\right)^2$ or $h' = \frac{1}{25}h$

Position of the ball above the ground

= h - h' = $h - \frac{h}{25} = \frac{24h}{25}$ m

27.

(b) 9.8

Explanation: W = m $\times$ g = m $\times$ 9.8

$\frac{W}{m}$ = 9.8

28. **(a)** have the same acceleration

Explanation: Air is absent in vacuum so no change will take place in acceleration to a freely falling body irrespective of their nature or mass.

29.

(b) pressure

Explanation: pressure = $\frac{force}{area}$

So, if the area is more pressure exerted will be less and vice versa. This is the reason why the shoulder strap in school bags are broad. This helps in reducing the pressure exerted on the shoulders of the person and thus makes it more comfortable to carry around.

30.

(c) U = W

Explanation: Weight is a force acting on an object and always acts downwards. Upthrust is a force exerted by fluid in an upward direction. For an object to float both of these forces act simultaneously and eventually get balanced with each other.

31.

(c) equally

Explanation: Acceleration= $\frac{change\ in\ velocity}{time\ taken}$

Thus the change in velocity with respect to the time taken will be equal in each of the respective cases.

32.

(b) 1.6 $\times$ 10^4 N m^{-2}

Explanation: Density of oil (ρ_0) = 0.60 g/cc = 600 kg/m^3

Density of water(ρ_w) = 1 g/cm^3 = 1000 kg/m^3

Pressure due to water =$h\rho_{\mathrm{w}}g$ = 1 $\times$ 1000 $\times$ 10 = 10000 N m^{-2}

Pressure due to oil = $h\rho_0 g$ = 1 $\times$ 600 $\times$ 10 = 6000 N m^{-2}

Total pressure = 10000 + 6000 = 16000 N m^{-2} = 1.6 $\times$ 10^4 N m^{-2}

33.

(b) lighter will have more K.E.

Explanation: The lighter body will have more K.E because kinetic energy is inversely proportional to the mass of the body.

34.

(b) is least on equator

Explanation: the value of g on earth is maximum at poles and decreases as we go from poles to the equator and is minimum at the equator.

35. **(a)** Less

Explanation: Pressure is inversely proportional to cross- sectional area.

36.

(b) universal gravitational constant

Explanation: We know that the gravitational force F = G $\frac{Mm}{r^2}$

Where F= force between mass

m = mass of the body

G = gravitational constant

r = Distant between two mass

Given M= 1unit and r = 1 unit, we get

F = G $\frac{1\times1}{1^2}$

$\Rightarrow$ F = G

37. **(a)** = 0

Explanation: When a body floats in a liquid, upthrust acting on the body is equal to the weight of the body in accordance with the law of flotation. That is, the weight of the body is balanced by the upthrust. So, the apparent weight of the body is zero as the net force acting on the body is zero.

38. **(a)** increases till it attains the weight in air

Explanation: Increases till it attains the weight in air as-

The buoyant force exerted by a fluid exerts upwards on a body submerged in it,

F= ρV

F = ρVg

where ρ is the density of the fluid, V is the volume of the fluid displaced, and 'g' is the acceleration due to gravity.

The liquid pushes the solid upward and by Newton's third law, there is a force of equal magnitude acting downward on the liquid.

39.

(b) any body thrown vertically up against gravity.

Explanation: When an object is thrown vertically upwards at the maximum height the velocity is going to be zero, however, acceleration is not zero. If there is no velocity, there can not be acceleration.

40. **(a)** 100 N

Explanation: Weight = Mg = 10 $\times$ 10 = 100 N

Section B

41. **(a)** (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)

Explanation: The correct order of match is given as (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii). These are the formulae to measure each physical quantity respectively.

42.

(d) 1-A, 2-C, 3-B, 4-D

Explanation: (1) The SI unit of Density is kilogram per cubic metre (kg/m3) and the cgs unit of gram per cubic centimetre (g/cm^3)

(2) Relative density of a substance = Density of the substance/Density of water at 4°C As relative density is the ratio of densities, it has no units.

(3) The acceleration which is gained by an object because of the gravitational force is called its acceleration due to gravity. Its SI unit is m/s^2 .

(4) The SI unit of G is Nm^2kg^{-2}

43. **(a)** (a) - (i), (b) - (iii),(c) - (ii), (d) - (iv)

Explanation: The correct order of match is given as (a) - (i), (b) - (iii),(c) - (ii), (d) - (iv).

(a) and (c) - Applications of the quantities given.

(b) - Relation due to change in gravity.

(d) - another name for the quantity given.

44.

(c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Explanation:

i. Value of G is always constant.
ii. An object floats over water if its density is less than water.
iii. S.I unit of pressure is Pascal.
iv. S.I unit of weight is Newton.

45.

(c) (a) - (ii), (b) - (iv),(c) - (i), (d) - (iii)

Explanation:

i. Fluids are a subset of the phases of matter and include liquids, gases, plasmas, and to some extent, plastic solids.
ii. Mass is the quantity of matter an object
iii. The weight of an object is the gravitational force between the object and the Earth.
iv. Every object in the universe attracts every other object with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them. The force is along the line joining the centers of two objects.

46. **(a)** (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Explanation:

a. Mass is measured by a beam balance.
b. Weight is measured by a spring balance.
c. A lactometer is an instrument that is used to check for the purity of milk by measuring its density.
d. A hydrometer or areometer is an instrument that measures the specific gravity (relative density) of liquids.

47.

(b) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

a. The SI unit of Density is kilogram per cubic metre (kg/m^3) and the cgs unit of gram per cubic centimetre (g/cm^3)
b. Relative density of a substance = Density of the substance/Density of water at 4°C As relative density is the ratio of densities, it has no units.
c. The acceleration which is gained by an object because of the gravitational force is called its acceleration due to gravity. Its SI unit is m/s^2.
d. The SI unit of G is Nm^2kg^{-2}.

48.

(b) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Explanation:

i. The force applied on a surface in a direction perpendicular or normal to the surface is called thrust.
ii. $g = 9.8\ m/s^2$
iii. $gm = 1.63 m/s^2$
iv. The Sun's gravity pulls on the planets, just as Earth's gravity pulls down anything that is not held up by some other force and keeps you and me on the ground.

49.

(c) (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Explanation:

a. **Fluid:** Gasses and liquids are all fluids because the molecules that make them up are in constant motion.
b. **Mass:** The mass of a body is the quantity of matter it contains.
c. **Weight:** The weight of an object is the force with which it is attracted towards the earth.
d. **Newton's law of gravitation:** Law of Gravitation states that force of attraction by which an object attracts other objects is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

50.

(c) (a) - (iii), (b) - (ii),(c) - (iv), (d) - (i)

Explanation: The correct order of match is given as (a) - (iii), (b) - (ii),(c) - (iv), (d) - (i). These form a perfect pair of values of quantities and reasons.

51.

(c) (a) - (iv), (b) - (i),(c) - (iii), (d) - (ii)

Explanation:

i. The universal law of gravitation successfully explained several phenomena which were believed to be unconnected: the force that binds us to the earth; the motion of the moon around the earth; the motion of planets around the Sun; and the tides due to the moon and the Sun.
ii. The weight of an object is the gravitational force between the object and the Earth.
iii. The force acting (thrust) per unit area is called pressure.
iv. The density of a liquid is the ratio of the mass per unit volume of the material on the water mass per unit volume.

52.

(d) 1-C, 2-B, 3-D, 4-A

Explanation: Column A can be explained correctly by column B in this sequence.

53.

(d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation:

i. Archimede's principle is also used in designing ships and submarines. The floating of a big ship is based on the Archimedes' principle.
ii. The weight of the object on the moon is one-sixth of weight of an object on earth.
iii. The strap of the school bag made wider to Reduce pressure on the shoulder.

iv. Buoyancy is the upward force exerted by fluids over the surface area of contact of an object which is immersed in fluids.

54.

(c) 1-B, 2-D, 3-A, 4-C

Explanation: 1) The accepted value of G is 6.673×10−11Nm2 kg−2

2) Ice floats on water because density of ice is less than water.

3) The unit of pressure in the SI system is the pascal (Pa).

4) The unit of measurement for weight is of force, which in the International System of Units (SI) is the newton.

55.

(d) (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Explanation: The correct order of the given match is (a) - (iv), (b) - (i), (c) - (iii), (d) - (ii).

a. F = $\frac{G.M.m}{r^2}$

b. g = $\frac{GM}{R^2}$

c. Relative density = $\frac{\text{Density of substance}}{\text{Density of water}}$

d. Pressure = $\frac{\text{Force}}{\text{Area}}$

56.

(c) (a) - (i), (b) - (iii), (c) - (ii), (d) - (iv)

Explanation:

i. Universal law of gravitation applied for the motion of moon around the earth.

ii. The gravitational force between a mass and the Earth is the object's weight.

iii. The force acting (thrust) per unit area is called pressure.

iv. The mass per unit volume of a substance is called density.

57.

(c) (a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Explanation:

i. Acceleration due to gravity depends on the latitude of the place.

ii. The accepted value of G is 6.673 $\times$ 10^{-11} Nm^2 kg^{-2}

iii. When an object falls from any height under the influence of gravitational force only, it is known as free fall.

iv. One kg wt = 9.8 N

58.

(c) 1-D, 2-A, 3-C, 4-B

Explanation: Column B is paired in accordance with the instruments for the measurement of quantities in column A

Section C

59. **(a)** Decrease

Explanation: g is inversely proportional to the square of the distance between them.

60.

(b) More

Explanation: The area is inversely proportional to the pressure.

61.

(b) Nm^2/kg^2

Explanation: Universal gravitational constant is represented by ‘G’. S.I. unit of universal gravitational constant is Nm^2/kg^2. As G = $\frac{Fxr^2}{Mm}$.

62. **(a)** zero

Explanation: Freely Falling Objects are the objects falling due to the force of gravity. During Free fall as there is no surface to provide the Normal Reaction Force on the freely falling body, the falling body has the feeling of weightlessness.

63.
(d) more
Explanation: The hydrometer sinks deeper in low-density liquids such as kerosene, gasoline, and alcohol, and less deep in high-density liquids such as brine, milk, and acids.

64. **(a)** purity, milk
Explanation: A lactometer is an instrument that is used to check for the purity of milk by measuring its density.

65.
(c) $\frac{1}{r^2}$
Explanation: The force acting on two bodies is inversely proportional to the square of the distance between them.
Therefore, F $\propto \frac{1}{r^2}$.

66.
(d) Increase
Explanation: W = mg, weight of an object is directly proportional to acceleration due to gravity of an object.

67. **(a)** decreases
Explanation: In the case of contact forces between two objects as the area of contact between the two increases the pressure per square unit of measure will decrease.

Section D

68. **(a)** (A)
Explanation: Mass is an independent quantity but weight is dependent on gravity. As gravity changes, weight also changes.

69.
(c) (D)
Explanation: When a body is dropped freely from a height its initial velocity is zero and thrown vertically upward its final velocity becomes zero and acceleration due to gravity is negative in case of the body thrown vertically upward.

70. **(a)** A
Explanation: The value of g depends upon the mass of the planet and radius, acceleration due to gravity does not change due to air friction. The body moving in an upward direction will have gravitational force in a downward direction.

71.
(c) A and C
Explanation: The variation in gravity is caused due to change in the mass of earth (exactly considering the volume of earth) at that place where you are measuring it. With change in g weight also changes.

72.
(d) step D
Explanation: Just on removing, some drops of water will be on the metal. So, it will show excess mass.

73.
(d) A and C are correct
Explanation: Acceleration due to gravity does not remain the same everywhere and so the weight also because 'g' is variable with the radius and mass of the body.

74.
(b) A
Explanation: The value of g depends upon the mass of earth and the distance between the centre of the object and earth. As depth increases, the value of acceleration due to gravity falls.

Section E

75.
(b) F_1 and F_2 are equal
Explanation: F_1 and F_2 are equal as Newton's law of gravitation obey Newton's third law of motion, i.e., if an object exerts a force on another object, then the second object exerts an equal and opposite force on the first object.